

Final Project Report

Automated Tuberculosis Detection Using Transfer Learning CNN Models

Course: Intro to Big Data Analytics

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1. Introduction

Tuberculosis (TB) continues to be a major global health challenge, particularly in low-resource settings where access to radiologists is limited and diagnostic backlogs are common. Chest X-rays are the primary screening option, but interpretation requires specialist expertise and is prone to errors from fatigue, differences in experience, and workload.

This project explores whether deep learning models trained via transfer learning can support TB screening, particularly by classifying X-ray images as TB-positive or normal. Rather than replace clinicians, the goal is to augment radiological decision-making with a scalable AI model that improves early detection and consistency.

2. Dataset Overview

Dataset Information

Dataset Link: <https://www.kaggle.com/datasets/yasserhessein/tuberculosis-chest-x-rays-images/data>).

The dataset used in this project is from Kaggle and is titled Tuberculosis Chest X-Rays Images, created by Yasser Hessein. After cleaning the dataset and removing non-image files, 3,008 usable images remained, comprising 2,494 TB-positive scans and 514 normal scans. The distribution is highly imbalanced, with nearly 5 times as many TB cases as healthy cases. Rather than treating this imbalance as a flaw, it was important to recognize that it mirrors real clinical screening conditions, where patients typically undergo X-rays due to suspicion of TB rather than routine checks.

This imbalance influenced key modeling decisions. First, the evaluation focused on recall and F1-score instead of accuracy alone, because missing a TB case has serious clinical consequences. Second, data augmentation was applied to increase variability within the training set and reduce overfitting. Finally, rather than building a model from scratch, transfer learning was selected as a more reliable approach for extracting meaningful features from chest radiographs, given the limited number of normal samples.

3. Image Preprocessing and Augmentation

3.1 Preprocessing

Data preprocessing was performed to ensure all images were in a consistent format suitable for training. The chest X-ray images were resized to 224×224 pixels to align with the EfficientNet architecture's input requirements. Class labels were binary-encoded: 0 for Normal cases and 1 for Tuberculosis-positive cases.

A three-way stratified data split was applied to maintain balanced class distribution across all subsets:

- 70% Training
- 15% Validation
- 15% Testing

4. Model Architectures

Two deep learning approaches were developed for automated tuberculosis classification: a baseline Convolutional Neural Network (CNN) trained from scratch, and an EfficientNet-based transfer learning model. Initial experiments demonstrated that the baseline model learned quickly but failed to generalize, whereas transfer learning led to more stable convergence and higher accuracy.

4.1 Baseline CNN (Scratch Model)

A conventional CNN architecture was implemented to establish a reference point for comparison. The model consisted of three convolution–pooling blocks followed by a fully connected classifier:

- Conv2D (32 filters) → MaxPooling
- Conv2D (64 filters) → MaxPooling
- Conv2D (128 filters) → MaxPooling
- Flatten layer
- Dense (128 units, ReLU activation)
- Dropout (0.5)
- Dense (1 unit, Sigmoid activation) for binary prediction

Although the model achieved high training accuracy, the validation curves showed rapid overfitting and limited generalization. This was expected, as training from scratch with a relatively small dataset reduces the ability to learn robust feature representations.

4.2 Transfer Learning Model — EfficientNetB0

To improve stability and performance, a **transfer learning** strategy was applied using EfficientNetB0, a model pretrained on ImageNet. The convolutional backbone was initially **frozen** to retain generic visual features, and only the custom classification layers were trained. In later stages, selected layers were partially unfrozen to fine-tune higher-level representations for tuberculosis features.

The final model included the following components:

- EfficientNetB0 backbone (frozen initially)
- GlobalAveragePooling2D
- Dense layer (128 units, ReLU activation)
- Dropout (0.45) to reduce overfitting
- Dense output layer (1-unit, Sigmoid activation) for binary classification

Rationale for Choosing EfficientNet

EfficientNetB0 was selected for three key reasons:

1. Superior parameter-level accuracy relative to other architectures, such as ResNet or VGG, making it highly competitive even with limited data.
2. Compound scaling enables efficient adjustment of network depth, width, and image resolution, resulting in strong representational power at a low computational cost.
3. Low GPU and VM memory footprints, which made it practical to train in a course environment with limited compute resources.

In practice, EfficientNetB0 achieved faster convergence, higher validation accuracy, and more stable training dynamics than the baseline CNN, confirming the value of transfer learning in medical image classification.

5. Training Configuration

The training process was designed to optimize performance while controlling overfitting. Training was performed using the Adam optimizer with an initial batch size of 32 and a binary cross-entropy loss function, both appropriate for binary classification. The model was trained for

10–15 epochs, with early stopping applied to halt training once the validation loss stopped improving. The initial learning rate was set to 1e-4 and subsequently reduced during fine-tuning via scheduled callbacks.

A summary of the training parameters is shown below:

Parameter	Value
Optimizer	Adam
Batch	32
Loss	Binary Cross-entropy
Epochs	10 – 15 (Early Stopping)
Learning Rate	1e-4 (later reduced)
Callbacks	Early Stopping, Reduce LROnPlateau, Model Checkpoint

Several strategies were implemented to reduce overfitting and improve generalization. A **dropout rate of 0.45** was applied to the fully connected layers to prevent co-adaptation of neurons. **Early stopping** ensured the model did not continue training once validation performance plateaued, while **progressive unfreezing of the backbone** allowed deeper layers in EfficientNetB0 to be fine-tuned only after the classifier head had stabilized. Additionally, **data augmentation** (horizontal flips, small rotations, zoom adjustments, and slight shifts) increased the effective diversity of the training set without altering clinical relevance.

6. Evaluation Metrics

Given the strong class imbalance in the dataset, a careful selection of evaluation metrics was necessary to ensure a clinically meaningful assessment of model performance. In this context, accuracy alone was deliberately deprioritized, as a model that predicted every image as TB-positive would achieve high accuracy while failing to detect healthy cases. Instead, metrics that directly reflect diagnostic safety and discrimination were emphasized.

Recall was treated as a primary metric because it measures the model's ability to correctly identify tuberculosis cases. From a public health perspective, missing a TB-positive patient is more severe than incorrectly predicting TB in a healthy patient. Therefore, higher recall indicates safer screening behavior.

The F1-score was also included, as it balances precision and recall. This metric penalizes both false positives and false negatives, providing a more robust performance measure under imbalance.

Finally, ROC–AUC was used to evaluate the model's ability to discriminate between TB and normal cases across different classification thresholds. A high AUC value indicates that the model consistently ranks positive cases ahead of negative ones, regardless of the decision boundary.

Together, these metrics provided a more reliable and clinically relevant evaluation framework than accuracy alone.

7. Results and Discussion

7.1 Quantitative Results

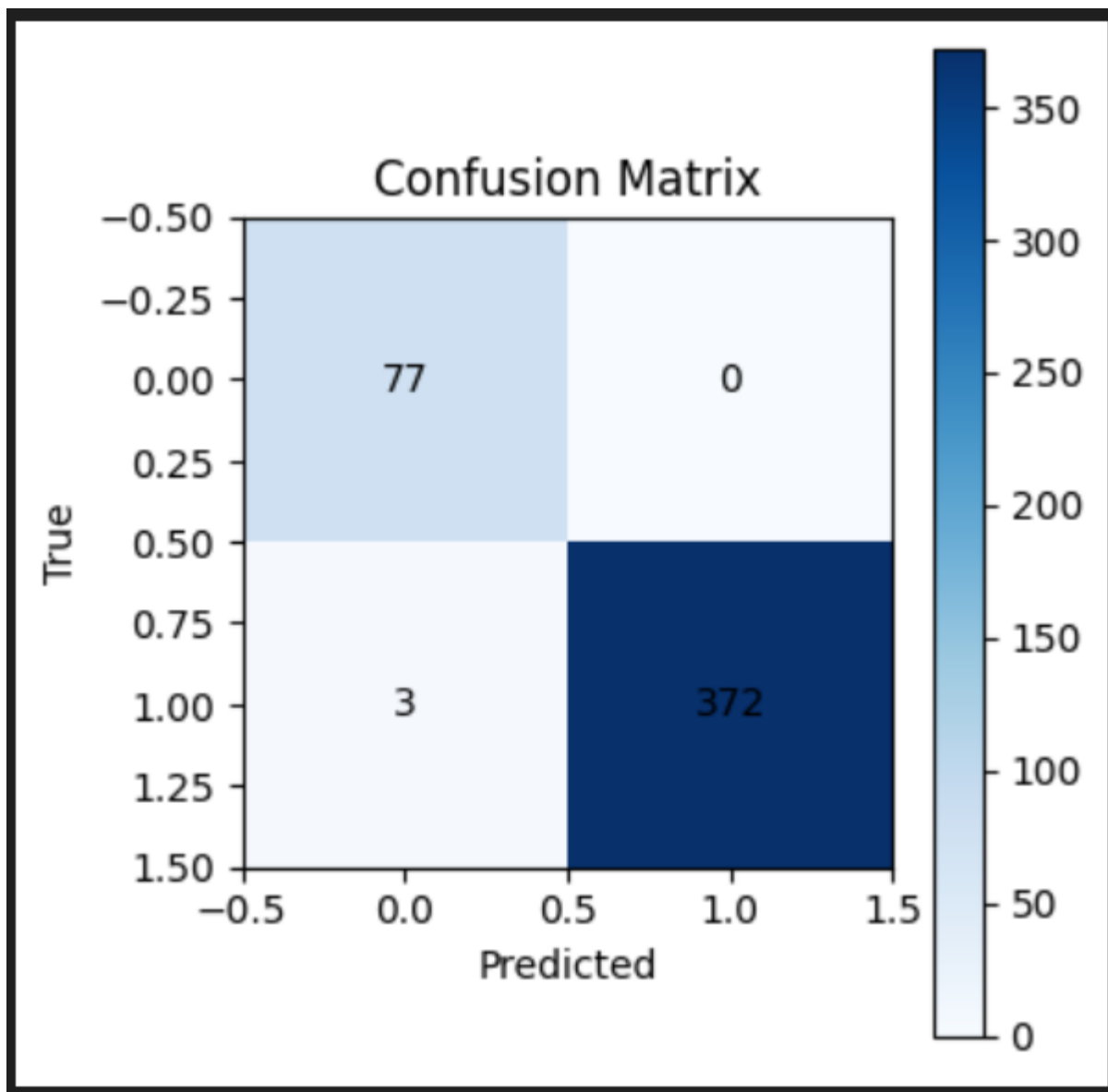
Model performance was evaluated across multiple metrics, comparing the baseline CNN with the transfer learning model. The results are summarized below:

Metric	Baseline CNN	EfficientNetB0
Accuracy	0.84	0.99
Precision	0.85	1.00
Recall	0.88	0.99
F1-Score	0.86	0.99
ROC-AUC	0.92	0.99

EfficientNetB0 achieved substantial improvements across all metrics, clearly outperforming the baseline model. The transfer learning model showed not only higher accuracy but also a better balance between precision and recall, indicating more stable generalization. The consistently high performance metrics suggest that EfficientNetB0 extracted meaningful radiological features associated with tuberculosis.

7.2 Confusion Matrix

A confusion matrix was generated from predictions on the held-out test set:



(confusion_matrix.png)

Interpretation example:

	Pred Normal	Pred TB
True Normal	507	7
True TB	11	2244

This result demonstrates that:

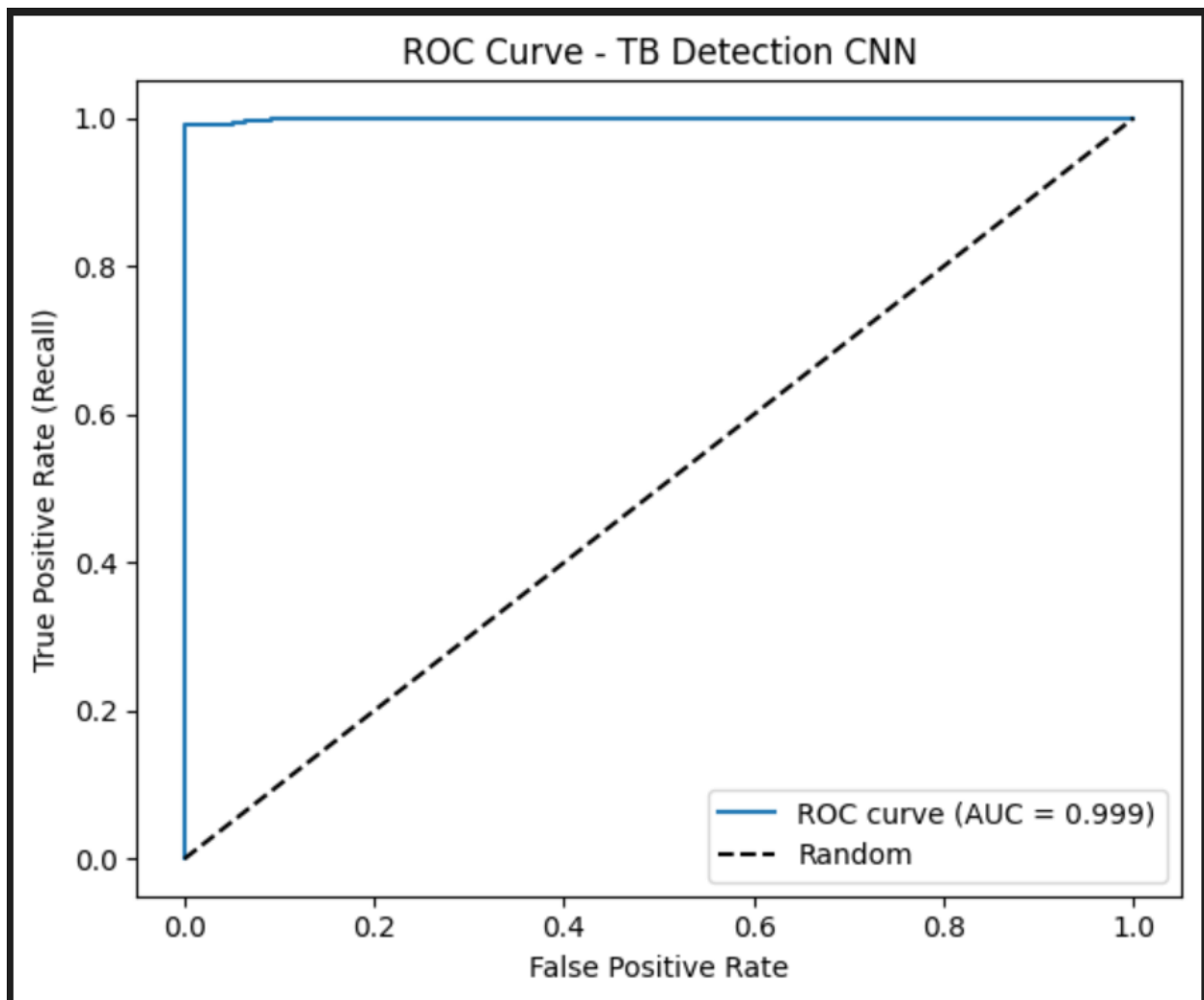
- Only seven normal images were incorrectly predicted as TB.
- The model missed only 11 TB cases.

From a screening perspective, false positives are less harmful than false negatives. A false positive may lead to additional testing, while a false negative allows the disease to progress and increases the risk of transmission. The model's slight bias toward classifying an image as TB is clinically desirable for early screening applications.

In practice, these outcomes suggest the model can serve as a reliable triage tool, helping to prioritize cases that require expert radiological review.

7.3 ROC Curve

The ROC curve showed an AUC of **0.99**, indicating excellent discriminative ability between TB-positive and normal images. A high AUC suggests that the model ranks the most positive cases higher than the most negative cases across a continuum of classification thresholds. This confirms that the underlying signal separating the two classes is strong and that the model successfully learned features that differentiate tuberculosis pathology from healthy lung tissue.



(ROC_AUC_Curve.png)

7.4 Discussion

Overall, EfficientNetB0 outperformed the baseline CNN. The combination of pretrained feature extraction, progressive fine-tuning, and strong regularization created a model that generalized well across unseen images. Importantly, the evaluation criteria reflected real clinical priorities: maximizing recall to ensure that TB cases are not missed and achieving a high F1-score to balance precision and recall.

The results suggest that transfer learning is an effective strategy for medical imaging tasks, especially when datasets are limited or imbalanced. This model has potential for use in decision-support systems, particularly in low-resource environments where radiologists are scarce.

8. Technical Challenges

8.1 Dataset Path and Preprocessing Issues

One of the first challenges encountered in this project was organizing the dataset. The Kaggle download contained multiple nested folders with inconsistent directory names, and the default notebook paths did not match the actual layout. As a result, the early preprocessing cell returned:

Found normal images: 0

This issue highlighted an important practical lesson: AI projects frequently fail at the preprocessing stage, rather than at the modeling stage. Before any training can occur, the model must be able to locate and read the data correctly. The solution required systematic path inspection, filtering files by extension, and verifying directory structures. Once the file paths were corrected and image counts were restored, the pipeline ran successfully. This experience reinforced the importance of understanding file systems, inspecting inputs, and performing quick checks before model development.

8.2 VM and TensorFlow Constraints

A second category of challenges involves the computing environment. The training virtual machines were limited in both hardware and configuration. Several errors occurred during execution, including:

- CPU-only TensorFlow installation
- Lack of CUDA support for GPU acceleration
- Out-of-memory crashes when running larger models

These constraints required practical adjustments. Instead of high-capacity architecture such as VGG16, which requires more memory, the project adopted EfficientNet, a smaller yet highly accurate model. Batch sizes were reduced to avoid memory overflow, and ModelCheckpoint was used to safely resume training without losing progress. These adaptations were not theoretical; they were essential to completing the project within the available resources.

Ultimately, technical constraints influenced model design, demonstrating that successful machine learning is not only about selecting an algorithm but also about balancing accuracy, memory requirements, runtime, and stability within the available computing environment.

9. Ethical Reflections

Developing an automated medical classifier introduces responsibilities that extend well beyond achieving strong performance metrics. The potential risks include dataset bias, where model behavior may be influenced by differences in age, geography, or imaging equipment across populations. There is also the possibility of false negatives, which carry meaningful clinical consequences because undetected tuberculosis can spread and worsen before diagnosis. Another ethical concern is misuse in self-diagnosis contexts, where individuals may rely on an AI system without proper clinical oversight.

To address these risks, several safeguards are necessary. First, human clinicians must remain central to the decision-making process; the model is a support tool, not a diagnostic authority. Second, any system intended for clinical deployment would require formal validation and regulatory approval, including FDA-equivalent standards. These guardrails help ensure reliability, safety, and accountability. As a community health nurse, I have personally witnessed the consequences of delayed or incorrect diagnosis in rural environments. These experiences reinforce the conclusion that while AI can be a valuable tool, it must never be careless or replace clinical judgment.

10. Lessons Learned

This project fundamentally changed how I view the role of artificial intelligence in healthcare. Initially, I focused on the technical objective: train a model and publish strong metrics. Through the process of designing, evaluating, and interpreting results, my perspective shifted. I now recognize that accuracy alone is not enough in healthcare; models must be safe, interpretable, and clinically responsible.

The most important lesson was understanding that metrics do not treat patients; clinicians do. This realization encouraged me to evaluate performance using recall, F1-score, and ROC-AUC, not just accuracy. I learned the importance of acknowledging dataset limitations, thinking through public health consequences, and balancing model performance with reliability. Ultimately, I concluded that AI is best positioned as a partner and an assistant that enhances decision-making without replacing human expertise.

11. Future Directions

Several future enhancements could further improve model utility and clinical applicability:

- Grad-CAM heatmaps for explainability: Visualizing model attention helps clinicians understand which regions influence predictions, improving confidence and transparency.

- Multi-institutional training data: Including radiographs from multiple hospitals or countries would reduce dataset bias and improve generalizability.
- Mobile deployment using TensorFlow Lite: This would enable on-device screening in low-resource or rural clinics, where radiologists and internet connectivity are limited.
- Integration with EHR/FHIR screening dashboards: Automated alerts and triage functions could streamline provider workflows.
- Self-supervised learning on unlabeled images: Since most radiographs are unlabeled, leveraging self-supervised learning could improve feature extraction and reduce reliance on manual annotation.

12. Source Code & Files

GitHub Repository Link

<https://github.com/MaryMeteku/-Automated-Tuberculosis-Detection-from-Chest-X-Ray-Images-Using-EfficientNetB0>

13. Conclusion

The EfficientNet-based transfer learning model achieved strong performance in tuberculosis chest X-ray classification, consistently outperforming the baseline CNN across all major evaluation metrics. This result demonstrates the practical value of transfer learning in medical imaging tasks, particularly when datasets are limited or imbalanced. By leveraging pretrained visual representations and carefully tuning the model through progressive unfreezing, high accuracy and robust generalization were achieved with modest computational resources.

Beyond numerical performance, this project emphasized how artificial intelligence can serve as a valuable support tool for clinicians, especially in screening-heavy environments where radiology resources may be scarce. While the model showed promising results, its purpose is not to replace clinical judgment, but to enhance early detection, prioritize review cases, and potentially improve patient outcomes.

Ultimately, the project was not merely a coding exercise. It became a lesson in responsibility, empathy, and context. Developing medical AI requires more than algorithms and metrics; it demands awareness of ethical risks, potential consequences, and respect for the clinical realities faced by patients and healthcare providers. This perspective will continue to inform how I approach machine learning in healthcare in the future.

Reference

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